

Overview of Feature selection

Subject: Feature selection

1.

$$\mathrm{mRMR} = \max_{S} \left[\frac{1}{|S|} \sum_{f_i \in S} I(f_i; c) - \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \right].$$

2.

Overview on metaheuristics methods A metaheuristic is a general description of an algorithm dedicated to solve difficult (typically NP-hard problem) optimization problems for which there is no classical solving methods. Generally, a metaheuristic is a stochastic algorithm tending to reach a global optimum. There are many metaheuristics, from a simple local search to a complex global search algorithm.

3.

Optimality criteria The choice of optimality criteria is difficult as there are multiple objectives in a feature selection task. Many common criteria incorporate a measure of accuracy, penalised by the number of features selected. Examples include Akaike information criterion (AIC) and Mallows's Cp, which have a penalty of 2 for each added feature. AIC is based on information theory, and is effectively derived via the maximum entropy principle. Other criteria are Bayesian information criterion (BIC), which uses a penalty of $\log \sqrt{n}$ for each added feature, minimum description length (MDL) which asymptotically uses $\log \sqrt{2 \log p}$, Bonferroni / RIC which use $\sqrt{2 \log p}$, maximum dependency feature selection, and a variety of new criteria that are motivated by false discovery rate (FDR), which use something close to $\sqrt{2 \log \frac{p}{q}}$. A maximum entropy rate criterion may also be used to select the most relevant subset of features.

4.

$$\mathrm{HSIC}_{\text{Lasso}} : \min_{\mathbf{x}} \frac{1}{2} \sum_{k,l=1}^n x_k x_l \mathrm{HSIC}(f_k, f_l) - \sum_{k=1}^n x_k \mathrm{HSIC}(f_k, c) + \lambda \|\mathbf{x}\|_1, \text{ s.t. } x_1, \dots, x_n \geq 0,$$

5.

Introduction A feature selection algorithm can be seen as the combination of a search technique for proposing new feature subsets, along with an evaluation measure which scores the different feature subsets. The simplest algorithm is to test each possible subset of features finding the one which minimizes the error rate. This is an exhaustive search of the space, and is computationally intractable for all but the smallest of feature sets. The choice of evaluation metric heavily influences the algorithm, and it is these evaluation metrics which distinguish between the three main categories of feature selection algorithms: wrappers, filters and embedded methods.

6.

The r_{cf_i} and $r_{f_i f_j}$ variables are referred to as correlations, but are not necessarily Pearson's correlation coefficient or Spearman's ρ . Hall's dissertation uses neither of these, but uses three different measures of relatedness, minimum description length (MDL), symmetrical uncertainty, and relief. Let x_i be the set membership indicator function for feature f_i ; then the above can be rewritten as an optimization problem:

7.

Here, $\overline{r_{cf}}$ is the average value of all feature-classification correlations, and $\overline{r_{ff}}$ is the average value of all feature-feature correlations. The CFS criterion is defined as follows:

8.

The mRMR algorithm is an approximation of the theoretically optimal maximum-dependency feature selection algorithm that maximizes the mutual information between the joint distribution of the selected features and the classification variable. As mRMR approximates the combinatorial estimation problem with a series of much smaller problems, each of which only involves two variables, it thus uses pairwise joint probabilities which are more robust. In certain situations the algorithm may underestimate the usefulness of features as it has no way to measure interactions between features which can increase relevancy. This can lead to poor performance when the features are individually useless, but are useful when combined (a pathological case is found when the class is a parity function of the features). Overall the algorithm is more efficient (in terms of the amount of data required) than the theoretically optimal max-dependency selection, yet produces a feature set with little pairwise redundancy. mRMR is an instance of a large class of filter methods which trade off between relevancy and redundancy in different ways.

9.

Quadratic programming feature selection mRMR is a typical example of an incremental greedy strategy for feature selection: once a feature has been selected, it cannot be deselected at a later stage. While mRMR could be optimized using floating search to reduce some features, it might also be reformulated as a global quadratic programming optimization problem as follows:

10.

where $F_{n \times 1} = [I(f_1; c), \dots, I(f_n; c)]^T$ is the vector of feature relevancy assuming there are n features in total, $H_{n \times n} = [I(f_i; f_j)]_{i,j=1 \dots n}$ is the matrix of feature pairwise redundancy, and $\mathbf{x}_{n \times 1}$ represents relative feature weights. QPFS is solved via quadratic programming. It is recently shown that QPFS is biased towards features with smaller entropy, due to its placement of the feature self redundancy term $I(f_i; f_i)$ on the diagonal of H .

11.

Application of feature selection metaheuristics This is a survey of the application of feature selection metaheuristics lately used in the literature. This survey was realized by J. Hammon in her 2013 thesis.

12.

Filter type methods select variables regardless of the model. They are based only on general features like the correlation with the variable to predict. Filter methods suppress the least interesting variables. The other variables will be part of a classification or a regression model used to classify or to predict data. These methods are particularly effective in computation time and robust to overfitting. Filter methods tend to select redundant variables when they do not consider the relationships between variables. However, more elaborate features try to minimize this problem by removing variables highly correlated to each other, such as the Fast Correlation Based Filter (FCBF) algorithm.